BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 10 — Acids, Bases and Salts

ANSWER KEY (Version 1)

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. According to Arrhenius theory, an acid produces: √ **B — H^+ ions** **1 mark**

The textbook (Section 10.1.1) states: "**An acid is a substance that ionizes in water to produce H^+ ions.**" NaOH (a base) produces OH^- ions (option A). Options C and D are wrong because Na^+ and O^{2-} are not the defining ions of an Arrhenius acid.

Q2. According to Bronsted-Lowry theory, a base: √ **C — accepts a proton** **1 mark**

The Bronsted-Lowry theory (Section 10.1.2) defines a **base as a proton acceptor**. Option A describes an acid (proton donor). Options B and D are Arrhenius descriptions, not Bronsted-Lowry. The Bronsted-Lowry theory was proposed in 1923 to overcome limitations of the Arrhenius theory.

Q3. Which cannot be classified as an Arrhenius acid? √ **C — CO_2** **1 mark**

CO_2 has **no H atoms** and cannot ionize in water to produce H^+ directly; it is therefore not an Arrhenius acid. This is from Review Question (i) in the textbook. HNO_3, H_2CO_3, and H_2SO_4 all contain hydrogen and ionize to yield H^+ ions in water. This limitation of the Arrhenius theory was one reason Bronsted and Lowry proposed their broader concept.

Q4. In $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, H_2O acts as a Bronsted-Lowry: √ **B — base** **1 mark**

H_2O **accepts a proton** from HCl to form the hydronium ion H_3O^+. A proton acceptor is a **Bronsted-Lowry base**. HCl donates the proton, so HCl is the Bronsted acid. This is Example 10.1 in the textbook.

Q5. An acid that ionizes completely in aqueous solution is a: √ **C — strong acid** **1 mark**

Section 10.2.1: "**An acid that ionizes completely in aqueous solution is called a strong acid.**" Weak acid ionizes only partially. Dilute vs. concentrated refers to concentration, not the degree of ionization, which is a common student confusion. HCl, HNO_3, and H_2SO_4 are all strong acids.

Q6. Ethanoic acid ionizes only 5% in water. It is a: √ **B — weak acid** **1 mark**

Section 10.2.1: "**Acids that do not ionize completely in aqueous solutions are called weak acids.**" Ethanoic acid (acetic acid, found in vinegar) ionizes only up to 5%. Its partial ionization is shown as a reversible arrow: $CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$. This is the classic textbook example of a weak acid.

Q7. Zinc + hydrochloric acid gives: √ **B — $ZnCl_2 + H_2$** **1 mark**

Section 10.3, property 3: Metal + Acid → Salt + Hydrogen gas. $Zn + 2HCl \rightarrow$ **$ZnCl_2 + H_2$**. Option A is wrong (ZnO would require oxygen, not acid). Option C is wrong (Cl_2 is not a product of acid-metal reactions). Option D is also wrong for the same reason.

Q8. $Mg(OH)_2$ + HCl neutralization; salt formed is: √ **C — $MgCl_2$** **1 mark**

This is from Review Question (iii). $Mg(OH)_2 + 2HCl \rightarrow$ **$MgCl_2$** $+ 2H_2O$. The metal from the base (Mg) combines with the anion from the acid (Cl^-). $MgSO_4$ would form if H_2SO_4 were used instead of HCl. MgO is the oxide, not a neutralization product.

Q9. A base soluble in water is called an: √ **C — alkali** **1 mark**

Section 10.2.2 (Alkalis): "**A base which is soluble in water is called an alkali.**" All alkalis are bases but not all bases are alkalis (e.g. $Cu(OH)_2$ is a base but not an alkali because it does not dissolve in water). NaOH and KOH are both alkalis.

Q10. Ammonia is a Bronsted-Lowry base because it: √ **C — can accept a proton** **1 mark**

Review Question (iv) and Section 10.1.2: NH_3 has a lone pair of electrons on N and accepts a proton from H_2O to form NH_4^+. It has **no OH^- group**, so options A and B (Arrhenius-based reasoning) are wrong. Option D (donating a proton) describes an acid, not a base.

Q11. Which is an amphoteric oxide? √ **C — ZnO** **1 mark**

Section 10.5: "**Zinc oxide (ZnO)** is an amphoteric oxide — it reacts with both acids and bases." Na_2O and CaO are basic oxides (they react only with acids). MgO is also a basic oxide. The other textbook examples of amphoteric oxides/hydroxides are Al_2O_3 and $Zn(OH)_2$.

Q12. SO_2 dissolved in rainwater produces: √ **C — H_2SO_3/H_2SO_4** **1 mark**

Section 10.3 (acid rain): SO_2 dissolves in water to form **sulphurous acid (H_2SO_3)**, which may be further oxidized to sulphuric acid (H_2SO_4). Nitric acid (HNO_3) forms from NO_x, not SO_2. HCl and H_3PO_4 are not components of acid rain from industrial sources.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) — Arrhenius definitions + ionization equations with state symbols **3 marks**

Arrhenius acid: A substance that ionizes in water to produce **H^+ ions**. **1 mark**
Arrhenius base: A substance that ionizes in water to produce **OH^- ions**. **1 mark**
Ionization equations (state symbols required):

```
HCl(g) ---H2O--> H+(aq) + Cl-(aq) NaOH(s) ---H2O--> Na+(aq) + OH-(aq)
```

1 mark (both equations correct with state symbols; (g) for HCl gas, (s) for solid NaOH, (aq) for all aqueous ions)

OR

Bronsted-Lowry acid: A substance that **donates a proton (H^+)** to another substance. **1 mark**
Bronsted-Lowry base: A substance that **accepts a proton (H^+)** from another substance. **1 mark**
In $HNO_3 + H_2O \rightarrow H_3O^+ + NO_3^-$: **$HNO_3$ is the Bronsted acid** (it donates H^+); **H_2O is the Bronsted base** (it accepts H^+ to form H_3O^+). **1 mark**

2(ii) — Strong and weak acids; HCl vs CH_3COOH **3 marks**

Strong acid: An acid that **ionizes completely** (100%) in aqueous solution. Examples: HCl, HNO_3, H_2SO_4. **1 mark**

Weak acid: An acid that **ionizes partially** (incompletely) in aqueous solution, establishing an equilibrium. Examples: CH_3COOH (ethanoic acid), H_2CO_3 (carbonic acid). **1 mark**

HCl is strong because all HCl molecules ionize in water: $HCl_{(g)} \rightarrow H^+_{(aq)} + Cl^-_{(aq)}$ (one-way arrow).

CH_3COOH is weak because only about 5% ionizes: $CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$ (equilibrium arrow shows most molecules stay un-ionized). **1 mark**

OR

Strong base: A base that ionizes completely in aqueous solution. Examples: NaOH, KOH, $Ca(OH)_2$. **1 mark**

Weak base: A base that ionizes to only a small extent in aqueous solution. Examples: NH_3 (ammonia), $Al(OH)_3$. **1 mark**

NaOH ionization equation with state symbols:

```
NaOH(s) ---H2O--> Na+(aq) + OH-(aq)
```

1 mark (state symbols required: (s) for solid NaOH; (aq) for both ions)

2(iii) — Ionic equation for HCl + NaOH neutralization; spectator ions; net ionic equation **3 marks**

Full ionic equation (charges and state symbols required):

```
Molecular: HCl(aq) + NaOH(aq) --> NaCl(aq) + H2O(l) Full ionic: H+(aq) + Cl-(aq) + Na+(aq) + OH-(aq) --> Na+(aq) + Cl-(aq) + H2O(l)
```

1 mark (all ions written with correct charges and (aq)/(l) state symbols)

Spectator ions: $Na^+_{(aq)}$ and $Cl^-_{(aq)}$ appear on both sides unchanged; they are spectator ions and are cancelled. **1 mark**

Net ionic equation:

```
H+(aq) + OH-(aq) --> H2O(l)
```

1 mark (This is the net ionic equation for all strong acid-strong base neutralizations.)

OR

$NH_4OH + HNO_3 \rightarrow NH_4NO_3 + H_2O$ **3 marks**

This is a neutralization reaction (acid + base → salt + water). Ammonium hydroxide (base) reacts with nitric acid to form ammonium nitrate (salt) and water.

```
NH4OH(aq) + HNO3(aq) --> NH4NO3(aq) + H2O(l)
```

Award 1 mark for correct reactants, 1 mark for correct products (NH_4NO_3 and H_2O), 1 mark for a balanced equation. This is Review Question 3 in the textbook. **3 marks**

2(iv) — Three characteristic properties of acids with equations **3 marks**

(Any three of the following; 1 mark each for property + correct equation.)

1. **Sour taste** — acids taste sour (e.g. citric acid in lemon juice). No equation required for this property. **1 mark**
2. **Turn blue litmus red** — acids change the colour of blue litmus paper to red. **1 mark**
3. **Reaction with metals:** $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ (salt + hydrogen gas). **1 mark**
4. **Reaction with metal carbonates:** $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$ (salt + water + carbon dioxide). **1 mark**
5. **Neutralization with bases:** $HCl + NaOH \rightarrow NaCl + H_2O$. **1 mark**

OR

(Any three of the following; 1 mark each for property + correct equation.)

1. **Bitter taste** — bases taste bitter (e.g. soap). **1 mark**
2. **Turn red litmus blue** — bases change the colour of red litmus paper to blue. **1 mark**
3. **Slippery touch** — aqueous solutions of bases have a slippery feel. **1 mark**
4. **Neutralize acids:** $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + H_2O$. **1 mark**
5. **React with ammonium salts:** $NaOH + NH_4Cl \rightarrow NaCl + NH_3 + H_2O$. **1 mark**

2(v) — Acids with metal carbonates (HCl/Na_2CO_3; $H_2SO_4/CaCO_3$) **3 marks**

General pattern: Metal carbonate + Acid → Salt + Water + Carbon dioxide. **1 mark**

```
Na2CO3 + 2HCl --> 2NaCl + H2O + CO2 [1 mark] CaCO3 + H2SO4 --> CaSO4 + H2O + CO2 [1 mark]
```

The CO_2 gas is observed as **bubbling**. The $CaCO_3$ reaction is important industrially (glass, paper, and soap production). Both reactions are from Section 10.3 of the textbook.

OR

General pattern: Metal + Acid → Salt + Hydrogen gas. **1 mark**

```
Zn + 2HCl --> ZnCl2 + H2 [1 mark] Mg + H2SO4 --> MgSO4 + H2 [1 mark]
```

The hydrogen gas is liberated as bubbles. These reactions are from Section 10.3, property 3. Note that the metal displaces hydrogen from the acid.

2(vi) — Alkalis: definition, examples, household products **3 marks**

Alkali: A base which is **soluble in water** is called an alkali. All alkalis are bases, but not all bases are alkalis. **1 mark**
Three examples from Table 10.3: NaOH (sodium hydroxide), KOH (potassium hydroxide), $Ca(OH)_2$ (calcium hydroxide). $Mg(OH)_2$ is also acceptable. **1 mark**
Household products containing alkalis: soaps, detergents, shampoos, toothpaste, drain cleaners, baking powder (soda). (Any two correct examples.) **1 mark**

OR

Bases react with ammonium salts to produce salt, ammonia gas, and water. The ammonia gas is released on heating. **1 mark**

```
NaOH + NH4Cl --> NaCl + NH3 + H2O [1 mark] KOH + NH4Cl --> KCl + NH3 + H2O [1 mark]
```

These reactions are from Section 10.4, property 5. The ammonia can be detected by its pungent smell or by turning moist red litmus paper blue.

2(vii) — Metallic oxides: formation and reaction with water 3 marks

Metallic oxides: Compounds formed by the reaction of metals with oxygen. They are basic in nature. 1 mark

```
(a) 4Na + O2 --> 2Na2O (sodium oxide) (b) 2Mg + O2 --> 2MgO (magnesium oxide)
```

1 mark (both equations correct)

When metallic oxides dissolve in water, they form metal hydroxides (bases):

```
Na2O + H2O --> 2NaOH MgO + H2O --> Mg(OH)2
```

The oxide ion (O^{2-}) accepts protons from water molecules, producing OH^- ions. 1 mark

OR

Amphoteric oxides: Metal oxides that react with **both acids and bases** are called amphoteric oxides. They show properties of both acids and bases depending on the reaction partner. 1 mark

Two examples of amphoteric oxides: ZnO (zinc oxide) and Al_2O_3 (aluminium oxide). 1 mark

Two examples of amphoteric hydroxides: $Zn(OH)_2$ (zinc hydroxide) and $Al(OH)_3$ (aluminium hydroxide). These do not dissolve in water but react with both acids and bases, hence the term amphoteric. 1 mark

2(viii) — H_2SO_4 ionization: Bronsted acid/base; strong or weak 3 marks

In the reaction $H_2SO_{4(l)} + H_2O \rightarrow 2H^+_{(aq)} + SO_4{}^{2-}{}_{(aq)}$:

H_2SO_4 is a Bronsted-Lowry acid because it **donates protons (H^+)** to water. H_2O acts as the Bronsted base by accepting those protons. 1 mark

H_2SO_4 **fully ionizes** in aqueous solution (one-way arrow, no equilibrium), meaning **all** its molecules release H^+ ions. 1 mark

H_2SO_4 is classified as a **strong acid** (Section 10.2.1). It is also a **diprotic (dibasic) acid** because each molecule can donate two H^+ ions, producing $2H^+$ and $SO_4{}^{2-}$. 1 mark

OR

In $H_2O + NH_3 \rightleftharpoons NH_4{}^+ + OH^-$:

H_2O is the Bronsted-Lowry acid (it donates a proton to NH_3). **NH_3 is the Bronsted-Lowry base** (it accepts the proton). 1 mark

Water is amphoteric because it can act as both a Bronsted acid (donates H^+) and a Bronsted base (accepts H^+). In $HCl + H_2O$, water acts as a base. In $H_2O + NH_3$, water acts as an acid. Substances that react with both acids and bases are called **amphoteric substances**. 2 marks

2(ix) — Five general properties of acids (Section 10.3) 3 marks

1. Acids have **sour taste**. ½
2. Acids change the colour of **blue litmus paper to red**. ½
3. Aqueous solutions of acids **conduct electricity** (they are electrolytes). ½
4. Acids react with most metals to produce a **salt and hydrogen gas**. ½
5. Acids react with bases (neutralization) to form a **salt and water**. ½

(An additional sixth property: acids react with metal carbonates to form salt, water, and CO_2.) ½

OR

1. Bases have **bitter taste**. ½
2. Bases change the colour of **red litmus paper to blue**. ½
3. Aqueous solutions of bases have a **slippery (soapy) touch**. ½
4. Bases react with acids (neutralization) to form **salt and water**. ½
5. Bases react with ammonium salts on heating to produce **salt, ammonia, and water**. ½

½ for correct general statement that these properties are from Table 10.1 (characteristic properties of bases).

2(x) — Acid rain: definition, gases, structural effects **3 marks**

Acid rain: Normal rain is slightly acidic due to dissolved CO_2. When **acidic oxide pollutants** from human activity (burning fossil fuels) enter the atmosphere, they dissolve in rainwater and make it significantly more acidic. This unusually acidic rainfall is called **acid rain**. **1 mark**

Main acidic gases responsible: sulphur dioxide (SO_2) and nitrogen oxides (NO_x). These are produced by burning fossil fuels and in vehicle exhaust. **1 mark**

Two damaging effects on structures:

(i) Acid rain corrodes metal structures such as bridges, railings, and ships because acids react with metals (Metal + Acid → Salt + H_2).

(ii) Acid rain dissolves limestone, marble, and concrete buildings and statues that contain $CaCO_3$: $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$. **1 mark**

OR

Why normal rain is slightly acidic: Atmospheric CO_2 dissolves in rainwater to form weak carbonic acid (H_2CO_3), giving normal rain a pH of about 5.6. No pollution is required for this mild acidity. **1 mark**

Contrast: Acid rain (from industrial SO_2/NO_x) has a pH typically below 5, much more acidic than normal rain. **1 mark**

Environmental consequences:

(i) Biological: acid rain kills aquatic organisms in lakes and rivers by lowering the pH of water, damaging ecosystems.

(ii) Structural: acid rain corrodes metal bridges and dissolves $CaCO_3$ in buildings and statues. **1 mark**

2(xi) — Comparison table: HCl, HNO_3, H_2SO_4 **3 marks**

Property	HCl (Hydrochloric acid)	HNO_3 (Nitric acid)	H_2SO_4 (Sulphuric acid)
Formula	HCl	HNO_3	H_2SO_4
Common use (one)	Cleaning metals, removing scale from boilers	Manufacture of fertilizers and explosives	Manufacture of chemicals, drugs, dyes, paints
Strong or weak	Strong acid	Strong acid	Strong acid

(1 mark per acid column — formula + use + classification correct.) **3 marks**

OR

Property	NaOH (Sodium hydroxide)	KOH (Potassium hydroxide)	$Ca(OH)_2$ (Calcium hydroxide)
Formula	NaOH	KOH	$Ca(OH)_2$
Common use (one)	Soap making, drain cleaners	Making liquid soap, shaving cream	Making mortar, plasters, cement
Strong or weak	Strong base	Strong base	Strong base

3 marks

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) — Arrhenius vs Bronsted-Lowry; classifying acids and bases in reactions **5 marks**

(a) Comparison of theories:

Aspect	Arrhenius Theory (1887)	Bronsted-Lowry Theory (1923)
Acid definition	Ionizes in water to give H^+ ions	Proton (H^+) donor
Base definition	Ionizes in water to give OH^- ions	Proton (H^+) acceptor
Applies to	Aqueous solutions only	Any solvent (broader scope)
Explains NH_3 as base?	No (no OH^- group)	Yes (NH_3 accepts a proton)

2 marks **Limitation overcome:** The Arrhenius theory cannot explain why NH_3 (ammonia) is a base because it contains no OH^- group and no H^+. The Bronsted-Lowry theory explains this because NH_3 accepts a proton from water, acting as a proton acceptor. **1 mark** **(b) Classification:**

(i) $HCl + H_2O \rightarrow H_3O^+ + Cl^-$: **HCl is the Bronsted acid** (it donates H^+ to water). H_2O is the Bronsted base (it accepts H^+). **1 mark**

(ii) $H_2O + NH_3 \rightleftharpoons NH_4^+ + OH^-$: **$NH_3$ is the Bronsted base** (it accepts H^+ from water to form NH_4^+). H_2O is the Bronsted acid here (it donates H^+). **1 mark**

OR

(a) Water as an amphoteric substance: Water can behave as both a Bronsted acid and a Bronsted base because it can both donate and accept protons. Such substances are called **amphoteric**. **1 mark**

Water as Bronsted BASE (accepts proton from HCl):

```
HCl + H2O --> H3O+ + Cl- (acid) (base)
```

Water as Bronsted ACID (donates proton to NH_3):

```
H2O + NH3 <==> NH4+ + OH- (acid) (base)
```

2 marks (1 mark per correct labelled equation)

(b) Classification from Exercise 7:

(i) $CH_3COOH_{(aq)} + H_2O \rightleftharpoons CH_3COO^-{}_{(aq)} + H_3O^+{}_{(aq)}$: **$CH_3COOH$ = Bronsted acid**; **H_2O = Bronsted base**. **1 mark**

(ii) $HCO_3^-{}_{(g)} + H_2O_{(l)} \rightleftharpoons CO_3^{2-}{}_{(aq)} + H_3O^+{}_{(aq)}$: **$HCO_3^-$ = Bronsted acid** (donates H^+); **H_2O = Bronsted base** (accepts H^+). **1 mark**

3(ii) — Characteristic properties of acids + ionic equation for neutralization 5 marks

(a) Three properties of acids with balanced equations: 3 marks

1. Reaction with metals (Metal + Acid → Salt + H_2):

```
Zn(s) + 2HCl(aq) --> ZnCl2(aq) + H2(g)
```

1 mark

2. Reaction with metal carbonates (Carbonate + Acid → Salt + H_2O + CO_2):

```
Na2CO3(s) + 2HCl(aq) --> 2NaCl(aq) + H2O(l) + CO2(g)
```

1 mark

3. Neutralization with bases (Acid + Base → Salt + Water):

```
HCl(aq) + NaOH(aq) --> NaCl(aq) + H2O(l)
```

1 mark **(b) Full ionic equation for HCl + NaOH neutralization:** 2 marks

Because HCl and NaOH are both strong electrolytes, they exist as fully dissociated ions in solution. NaCl also dissociates completely. H_2O is a pure liquid and is written as a molecule.

```
Full ionic equation (ALL charges and state symbols required): H+(aq) + Cl-(aq) + Na+(aq) + OH-(aq) --> Na+(aq) + Cl-(aq) + H2O(l) Spectator ions (appear unchanged on both sides): Na+(aq) and Cl-(aq) Cancel spectator ions to get the net ionic equation: Net ionic equation: H+(aq) + OH-(aq) --> H2O(l)
```

1 mark for full ionic equation with correct charges and state symbols. 1 mark for identifying spectator ions and writing the correct net ionic equation.

Why all strong acid-strong base neutralizations share the same net ionic equation: The only reaction occurring is between H^+ and OH^- to form water. The cation from the base and the anion from the acid are both spectator ions regardless of which strong acid and base are used.

OR

(a) Three properties of bases with equations: 3 marks

1. Neutralization with acids:

```
H2SO4(aq) + 2NaOH(aq) --> Na2SO4(aq) + 2H2O(l) [1 mark]
```

2. Reaction with ammonium salts:

```
NaOH(aq) + NH4Cl(s) --> NaCl(aq) + NH3(g) + H2O(l) [1 mark]
```

3. Red litmus turns blue; slippery touch (no equation needed for physical properties). 1 mark **(b) H_2SO_4 reactions:** 2 marks

```
(i) H2SO4(aq) + Mg(OH)2(aq) --> MgSO4(aq) + 2H2O(l) [1 mark] (ii) H2SO4(aq) + Na2CO3(s) --> Na2SO4(aq) + H2O(l) + CO2(g) [1 mark]
```

Three products in (ii): Na_2SO_4 (salt), H_2O (water), CO_2 (gas). Bubbling is observed because CO_2 gas is released when the carbonate is decomposed by the acid.

3(iii) — Oxides and hydroxides of metals; basic vs amphoteric; reactions with water **5 marks**

(a) Definitions and differences:

Metallic oxides: Compounds formed when metals react with oxygen. Most are **basic** in nature (they react with acids to form salt and water). Example: Na_2O, MgO, CaO. **1 mark**

Amphoteric oxides: Metal oxides that react with **both** acids and bases. They are neither purely acidic nor purely basic.

Aspect	Basic oxide	Amphoteric oxide
Reacts with acid?	Yes	Yes
Reacts with base?	No	Yes
Two examples	Na_2O, MgO	ZnO, Al_2O_3

2 marks **(b) Reaction with water:** **2 marks**

```
(i) Na2O(s) + H2O(l) --> 2NaOH(aq) [1 mark] Product: NaOH = sodium hydroxide = STRONG base. Use: soap making, drain cleaners. (ii) CaO(s) + H2O(l) --> Ca(OH)2(aq) [1 mark] Product: Ca(OH)2 = calcium hydroxide (slaked lime) = STRONG base. Use: making mortar, plasters, cement.
```

OR

(a) Why metallic oxides are basic: Metallic oxides are ionic compounds containing the oxide ion O^{2-}. When they dissolve in water, the O^{2-} ion immediately accepts protons from water (acting as a Bronsted base):

```
O2-(aq) + H-OH(l) --> 2OH-(aq)
```

This produces OH^- ions, giving the solution its basic character. Most metallic oxides show characteristic properties of bases (react with acids, turn red litmus blue). **3 marks** **(b) Water as amphoteric; other examples:** **2 marks**

Amphoteric substance: Reacts with both acids and bases. Water is the classic example.

From Section 10.1.2: Water acts as **base** with HCl (accepts H^+); water acts as **acid** with NH_3 (donates H^+). **1 mark**

One other **amphoteric oxide**: ZnO or Al_2O_3.

One other **amphoteric hydroxide**: $Zn(OH)_2$ or $Al(OH)_3$. **1 mark**

3(iv) — Acid rain: definition, gases, formation, damaging effects **5 marks**

(a) Definition of acid rain: Normal rain has a pH of about 5.6 because CO_2 dissolves in it to form weak carbonic acid. As a result of human activity, **acid-forming gases** enter the atmosphere and make rainwater significantly more acidic (pH below 5). This unusually acidic rain is called **acid rain**. **1 mark**

Main acidic gases and how they form acids in rainwater:

- **Sulphur dioxide (SO_2)**: from burning coal, oil, and in factory emissions. Dissolves in water to form sulphurous acid (H_2SO_3), which oxidizes further to sulphuric acid (H_2SO_4). **1 mark**
- **Nitrogen oxides (NO_x)**: from vehicle exhausts and industrial combustion. Dissolve in water to form nitric acid (HNO_3). Acid rain thus contains dissolved HNO_3 and H_2SO_4. **1 mark**

(b) Four damaging effects of acid rain: **2 marks**

- (i) **Metal structures:** Acid rain corrodes metal bridges, railings, and pipelines because acids react with metals (Metal + Acid → Salt + H_2). **½**
- (ii) **Stone buildings and statues:** Structures made of limestone or marble ($CaCO_3$) dissolve in acid rain:

  ```
  CaCO3(s) + H2SO4(aq) --> CaSO4(aq) + H2O(l) + CO2(g)
  ```

  ½
- (iii) **Aquatic life:** Acid rain lowers the pH of rivers and lakes, making them too acidic for fish, amphibians, and other aquatic organisms to survive. **½**
- (iv) **Soil and vegetation:** Acid rain leaches important mineral nutrients from soil and damages plant roots, reducing crop yields and harming forests. **½**

OR

(a) Properties of acids that make acid rain corrosive: Acid rain inherits all the corrosive properties of acids described in Section 10.3 and Table 10.1:

- Acids react with metals: $Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$ (corrodes iron structures). **1 mark**
- Acids react with metal carbonates: $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$ (dissolves limestone buildings). **1 mark**
- Acids are corrosive to skin (Table 10.1 property 4). **1 mark**

Buildings and statues contain metals and $CaCO_3$ (marble, limestone), both of which react with the H_2SO_4 and HNO_3 in acid rain, causing irreversible structural damage. **(b) Normal rain vs acid rain; environmental consequences:** **2 marks**

Normal rain: CO_2 from the atmosphere dissolves in rainwater to form H_2CO_3 (a weak acid), giving normal rain a pH of about 5.6. This is natural and not considered pollution. **½**

Acid rain: Contains dissolved H_2SO_4 and HNO_3 from industrial SO_2 and NO_x. pH can fall below 5 or even below 4 near industrial areas. **½**

(i) **Water bodies:** Acid rain lowers the pH of lakes and rivers, killing fish and aquatic organisms, disrupting entire food chains. **½**

(ii) **Soil:** Acid rain leaches nutrients (calcium, magnesium) from soil and releases toxic aluminium ions, damaging plant roots and reducing agricultural productivity over time. **½**